

LANCE: An LLM Agent for Network Compromise Evaluation with IoTChainBench

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Abstract—LLM penetration-testing agents are typically evaluated on isolated hosts, while IoT/OT compromises often depend on reachability across gateways, brokers, and segmented services. This leaves a methodological gap: single-host corpora do not measure pivots, protocol roles, or evidence depth. We present IoTChainBench, a reproducible network-scale IoT benchmark with 23 public Proxmox/Ansible scenarios, 220 virtual-device instances, 266 injected vulnerabilities, 53 negative controls, and six additional sealed evaluation profiles. We also introduce LANCE (LLM Agent for Network Compromise Evaluation), a six-phase network-native harness for topology analysis, active discovery, triage, verification, intrusion attempts, and report generation. We introduce a fail-closed evaluation protocol that separates detection, verified exploitation, severity coverage, enforced network pivots, and logical dependencies. The confirmatory comparison uses blind execution on the six sealed profiles, three pre-committed repetitions per profile, equal scenario weighting, and zero credit for missing runs or unsupported claims. Public scenarios are reserved for development, ablation, and informed-versus-blind diagnostics.

Index Terms—IoT security, penetration testing, large language models, autonomous agents, benchmark, cyber-physical systems

1. Introduction

IoT/OT deployments are compromised through paths, not just devices. A camera, broker, gateway, or PLC may be low impact in isolation but high impact when it enables access to a segmented service. Yet LLM penetration-testing agents are still evaluated mostly on isolated hosts, containers, or CTF-style services scored independently. Such benchmarks cannot measure whether an agent discovers pivot-gated vulnerabilities, reasons over IoT/OT protocols such as MQTT, Modbus, CoAP, or LoRaWAN, or produces evidence before reporting a finding.

Contributions. This paper makes three contributions.

- **IoTChainBench:** a reproducible network-scale IoT benchmark with 23 public scenarios (220 virtual-device instances, 266 vulnerabilities, and 53 controls) plus six sealed profiles, with per-vulnerability ground truth and infrastructure-verified reachability.

- **LANCE:** a six-phase network-native LLM harness for discovery, triage, verification, intrusion, and reporting across multi-device IoT networks, with normalized finding schemas and evidence-level scoring.
- **Evaluation:** a pre-committed, blind, sealed protocol with three scores—Detection F1, Verified F1, and Verified Severity Coverage—plus separate controls, verified path coverage, network-pivot reach (MHR), dependency-hop reach (DHR), cost, and completeness.
All artifacts are released at ANONYMIZED_URL.

2. Related Work

2.1. LLM Penetration-Testing Agents

LLM penetration-testing agents have evolved from single-agent shell-driving systems to multi-agent and model-specialized pipelines. Early systems such as Happe and Cito [1] and PENTESTGPT [2] showed that LLMs can plan and execute attacks on isolated VMs and CTF-style targets. Later systems, including HACKINGBUDDYGPT [3], AUTOATTACKER [4], VULNBOT [5], AUTOPENTESTER [6], CAI [7], and XOFFENSE [8], improve tool use, multi-agent planning, retrieval, or model specialization.

Despite these advances, reported evaluations remain primarily single-target or challenge-level. They do not score subnet reachability, pivot depth, or per-vulnerability progress across IoT/OT protocols such as MQTT, Modbus, LoRaWAN, and CoAP. This is the gap addressed by IoTChainBench and LANCE.

2.2. LLM Cybersecurity Benchmarks

Existing LLM cybersecurity benchmarks mainly score challenge completion. AUTOPENBENCH [9] uses 33 Docker-container tasks with milestone and flag-based scoring, while VULHUB [10], CAIBENCH [11], NYU CTF BENCH [12], and CYBENCH [13] cover vulnerable services, CTF tasks, attack-and-defense ranges, or knowledge tests. These corpora are useful for measuring exploit progress on isolated targets, but they do not provide per-vulnerability ground truth tied to reachability, hop depth, or IoT/OT protocol roles.

Multi-host cyber ranges such as LUDUS [14] and GOAD [15] model lateral movement in enterprise Active

Directory networks, but they are training labs rather than scored LLM benchmarks. They do not ship an IoT/OT protocol taxonomy, per-vulnerability oracle, or evaluator. IoTChainBench fills this gap by combining deployable multi-device IoT networks with reachability-aware ground truth and evidence-level scoring.

2.3. Attack Graphs and Network Reachability

Classical attack-graph systems model how vulnerabilities compose across a network. Early graph-based analysis [16], model-checking approaches [17], and Datalog-based systems such as MULVAL [18] compute reachable attacker states from static host and service descriptions. LANCE reuses this reachability view but applies it to live LLM-driven assessment: edges are validated by probes, actions are selected by the agent, and the graph is updated as hosts and paths are discovered.

2.4. IoT Security Context

Scenario design and ground-truth taxonomy draw on the OWASP Internet of Things project’s IoT Top 10 categories [19], MITRE ATT&CK for ICS [20] for OT-tactic annotations, and NIST SP 800-82 Rev. 3 [21] for OT segmentation conventions.

Existing IoT security artifacts mainly target either traffic classification or single-device firmware analysis. Network-traffic datasets such as IoT-23 [22], TON_IoT [23], and Edge-IIoTset [24] provide labeled traces for intrusion detection, but not deployable pentest targets with controllable reachability. Firmware-focused work such as Costin *et al.* [25], FIRMADYNE [26], FIRMAE [27], and IoT-GOAT [28] exercises single-device analysis. IoTChainBench instead evaluates active assessment on deployed multi-device IoT networks where firewall rules enforce which services are directly reachable and which require a pivot, with per-vulnerability ground truth.

3. Threat Model and Problem Definition

Problem. Given a target subnet and an entry point e , network-scale penetration testing requires discovering reachable hosts, identifying vulnerabilities, producing evidence of exploitability, and reporting findings across the full network. The unit of evaluation is the network rather than an individual host: some high-impact IoT/OT vulnerabilities are directly exposed, while others become reachable only through gateway or zone-transition pivots. Single-host metrics cannot capture this reachability distinction.

Attacker model. The attacker has Layer-3 access to the target subnet but no prior credentials, out-of-band software inventory, or privileged vantage point. They may use standard pentest tools (Nmap, MQTT clients, SSH, HTTP) through an LLM tool-calling agent under a bounded turn budget. Their objective is data exfiltration or access to OT services behind segmentation boundaries.

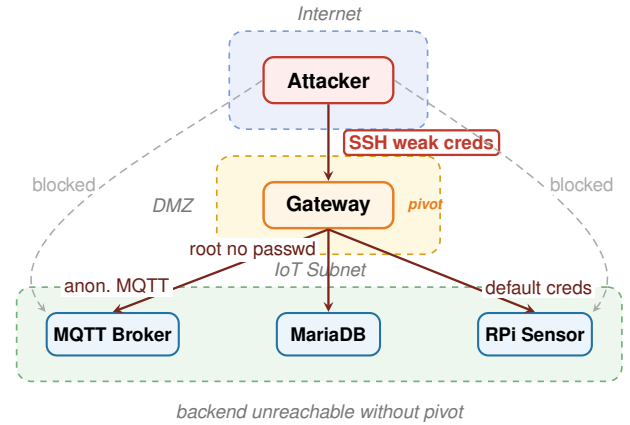


Figure 1. Conceptual pivot-gated attack surface. This diagram is not Scenario S3, which is a flat network. The confirmatory public multi-hop fixture is S20, where two isolated VLANs require successive entry and relay pivots and direct-access controls are verified before each run.

Evidence levels. A finding is credited at the deepest level the agent demonstrates: L1 detection, where a scan or banner confirms an issue; L2 exploitation, where active interaction demonstrates it; and L3 exfiltration, where sensitive content is retrieved. These levels guide LANCE Phase 4 and evaluator scoring.

Scope. We evaluate IP/application-layer attacks on virtual IoT networks. Physical, side-channel, RF-layer, firmware-extraction, persistence, and insider attacks are out of scope.

Figure 1 illustrates the conceptual distinction between direct reachability and a pivot-gated service. The benchmark attributes positive MHR depth only when deployment-time reachability checks enforce that distinction.

4. IoTChainBench: A Network-Scale Benchmark

IoTChainBench is designed to evaluate network-scale IoT/OT assessment rather than isolated-host exploitation. Scenarios are deployed reproducibly from Ansible playbooks on Proxmox and vulnerabilities are injected idempotently. Flat application workflows, logical prerequisite chains, and enforced network pivots are represented separately. In particular, a scenario receives positive network-pivot depth only when routing or firewall checks demonstrate that the target is unreachable from the original evaluator vantage point. Scenarios mix IoT/OT protocols such as MQTT, Modbus, OPC UA, BACnet, LoRaWAN, CoAP, SNMP, SSH, HTTP, FTP, and Telnet, with per-vulnerability YAML ground truth recording device, IP, severity, category, structural matching contract, indicators, and verification commands.

Design goals. The benchmark is built around three requirements:

- **Deployability:** every scenario is generated from Ansible and Proxmox artifacts, so runs can be recreated rather than approximated from static descriptions.
 - **Reachability awareness:** firewall rules encode which services are exposed from the entry point and which become reachable only after a gateway or zone transition.
 - **Scorable evidence:** the oracle records expected indicators and verification commands, allowing the evaluator to distinguish shallow detection from stronger exploit evidence.
- The public development split contains 23 scenarios (Table 1); six additional profiles are evaluated only by the sealed controller.

TABLE 1. IOTCHAINBENCH SCENARIO CATALOGUE.

IDs	Role in protocol	Dev	Vulns	Controls
S1--S12	Legacy public development and scale	116	209	0
S13	VLAN segmentation development fixture	17	20	0
S14	Sparse mixed-hardening controls	11	4	8
S15--S18	Stateful logical dependency chains	28	14	14
S19	Independent safe OT protocol checks	8	5	2
S20	Two enforced network pivots	5	4	3
S21--S22	Low prevalence and exploit diversity	26	6	22
S23	Wireless-to-firmware logical chain	9	4	4
Public total		220	266	53
S24--S29	Sealed confirmatory profiles	private until retirement		

4.1. Deterministic Vulnerability Injection

Vulnerabilities are injected by a dedicated Ansible role parameterized by `scenario_id`, making deployments reproducible and idempotent. Scenario YAML files declare router-level weaknesses (Telnet, WAN administration, FTP) and per-service roles that trigger targeted injections such as anonymous MQTT, weak SSH credentials, and passwordless database roots.

Across the public split, IoTChainBench injects 266 vulnerabilities and evaluates 53 explicit negative controls. Public S1–S23 are never used for a confirmatory leaderboard: they support implementation, ablation, matching audits, and informed-versus-blind diagnostics. S24–S29 are absent from the worker image and are instantiated with controller-side seeds.

4.2. Ground-Truth Schema

Each public scenario ships a YAML ground-truth file with one entry per injected vulnerability. Entries record the affected device, IP, role, severity, category, optional CVE, accepted semantic types, structural matching fields, expected indicators, verification command, and two distinct depth variables. `network_pivot_depth` counts enforced changes of network vantage point; `dependency_depth` counts ordered logical prerequisites. The historical `hop_depth` field is ignored by the v3.4 MHR scorer.

Scenarios may declare tightly bounded advisory types, but these are capped, deduplicated, and require traceable evidence. Hardened comparators are encoded as explicit negative controls and reported separately as a clean-run rate.

4.3. Scoring and Diagnostic Fields

Each LLM finding is matched one-to-one to ground truth under a versioned explicit contract. A match requires the target IP, an accepted semantic type (or a target-bound, offline-validated CVE), and the vulnerability-specific `required_dimensions`. Optional declared dimensions reject contradictions but are not all mandatory. There is no IP–severity fallback. Unmatched findings are false positives and duplicate claims are collapsed.

The main table reports three values. *Detection F1* measures correctly identified vulnerabilities whether or not exploitation succeeds. *Verified F1*, the primary score, counts a true positive only when a fresh Phase 4 tool record is target-compatible and its output semantically demonstrates the finding. *Verified Severity Coverage* is the fraction of total ground-truth severity weight (critical 4, high 3, medium 2, low 1) whose vulnerabilities are verified. Missing proof in a contract-compatible run receives zero verified credit.

For depth k , `MHR@k` is recall restricted to vulnerabilities with `network_pivot_depth ≥ k`; `DHR@k` applies the same definition to `dependency_depth`. Verified variants additionally require proof. Undefined depths are reported as N/A rather than zero. Attack-path coverage requires all referenced findings; verified path coverage also requires proof and an ordered Phase 5 chain.

5. LANCE: An LLM Agent for Network Compromise Evaluation

Existing LLM pentest harnesses typically process targets as a flat list, letting context grow with network size. LANCE instead decomposes network assessment into six phases with typed deliverables. Its key design choice is device- and vulnerability-level parallelism: constrained micro-agents analyze individual targets or findings, and their outputs are merged deterministically into a network-level state. This keeps early-stage contexts bounded as the number of devices grows, while cross-device reasoning is deferred to the intrusion and reporting phases. LLM-produced vulnerability types are canonicalized into the benchmark taxonomy before scoring, reducing noise and duplicate findings.

5.1. Pipeline

LANCE separates network assessment into discovery, local analysis, evidence collection, lateral movement, and reporting. **Phase 1** builds a directed topology graph $G = (V, E)$ whose vertices are devices and whose edges are protocol-labeled links. In informed mode, this graph is loaded from benchmark metadata; in blind mode, it starts empty and is populated by active discovery. **Phase 2** scans the target CIDR for hosts, services, versions, L2 vendors, and protocol endpoints (MQTT, SSH, HTTP). **Phase 3** performs per-device triage: each device receives a deterministic scanner pass and a constrained LLM micro-agent, whose outputs are merged and deduplicated. **Phase 4** verifies candidate findings with operational tools such as SSH, MQTT,

MySQL, Modbus, and Telnet, attempting the deepest evidence level achievable (L1–L3). **Phase 5** conducts intrusion and lateral movement over observed or asserted topology edges, recording each hop with source, destination, protocol, and credentials. **Phase 6** emits a network-scoped report with device inventory, validated findings, attack-surface summary, intrusion narrative, and remediation priorities.

5.2. Cost Envelope

For a network with n devices and v candidate findings per device on average, LANCE issues $O(nv)$ LLM invocations: four global calls (Phases 1, 2, 5, and 6), n per-device triage calls (Phase 3), and nv per-finding verification calls (Phase 4). Each invocation has a fixed turn budget (50 for Phases 1–2, 10 for Phases 3–4, 30 for Phase 5, and 20 for Phase 6), so cost scales with observed devices and findings rather than with combinations of network paths.

6. Experimental Setup

Infrastructure. All scenarios run as LXC containers on a Proxmox VE host, connected through a dedicated Linux bridge with an OpenWrt gateway. Deployment and teardown are fully automated with Ansible.

Model. For the architecture comparison, all LLM-driven systems use the same general-purpose model, **MiniMax-M2.7** [29], with fixed prompts and temperature 0 decoding. This controls for model choice while focusing the comparison on architecture rather than fine-tuning.

Baselines. We compare LANCE with two adapted LLM-agent baselines:

- a CAI-style single-agent tool loop [7];
- a VulnBot-style multi-agent pipeline with a penetration task graph [5].

The confirmatory comparison runs every system in blind mode from the same controller-issued network scope. Baseline adapters must emit the same target, type, structural fields, evidence references, and tool-call provenance as LANCE; the normalizer may not read ground truth or synthesize missing fields. Informed LANCE runs are restricted to the public diagnostic split and are not part of the sealed architecture claim.

Metrics and variance. We report Detection F1, Verified F1 (primary), Verified Severity Coverage, clean-run rate, verified path coverage, verified MHR/DHR, cost, and planned/completed run counts. The confirmatory matrix is $\{\text{LANCE, CAI, VulnBot}\} \times \text{S24–S29} \times \text{three paired controller repetitions}$. Repetitions are averaged within profile before a six-profile macro-average. A pre-committed but missing repetition contributes zero; no incomplete metric is silently omitted. We report paired effect sizes and 95% confidence intervals across controller-matched repetitions.

7. Evaluation

7.1. Pre-registered Confirmatory Matrix

The primary architecture comparison comprises 54 blind sealed runs: three systems (LANCE, CAI, and VulnBot), six controller-only profiles (S24–S29), and three paired repetitions. The suite manifest fixes the runner commit, model and provider, tool proxy, budgets, controller epoch, scoring policy, and planned run grid before the first execution. No prompt, adapter, or normalizer is changed after the first confirmatory result. A missing or crashed cell contributes zero to Detection F1, Verified F1, and Verified Severity Coverage; completeness is published with the score.

7.2. Public Diagnostics and Reporting

We separately execute S14–S23 in informed and blind mode with three repetitions per condition (60 runs). These public results diagnose topology-prior effects, matching errors, controls, exploit diversity, DHR, and the S20 network-pivot fixture, but do not support the sealed generalization claim. A 16-run smoke pilot on S20–S23 is used only to validate adapters and infrastructure and is excluded from all reported estimates.

The main result table contains, for each system, Detection F1, Verified F1, Verified Severity Coverage, clean-run rate, verified path coverage, verified MHR@1/2, cost, and completed/planned runs. We report paired differences against LANCE with 95% confidence intervals. Numerical cells are inserted only after the full sealed suite has completed and its aggregate signature has been verified. Consequently, the historical strict-v2 table below is retained in the source for provenance but deliberately excluded from this draft.

8. Discussion and Limitations

The revised design isolates three questions that the previous evaluation mixed.

- **Identification is not verification.** Detection F1 measures whether the system names the right vulnerability; Verified F1 determines whether the claim is supported by fresh target-bound execution evidence.
- **Logical depth is not network depth.** S15–S18 and S23 test ordered application prerequisites through DHR. Only infrastructure-enforced reachability transitions, such as S20, contribute to MHR.
- **Controls are not positive findings.** Hardened comparators produce a separate clean-run rate. Mixing them into a composite F1 obscures whether a failure comes from missed vulnerabilities or over-reporting.

Several limitations remain.

- **Virtualization boundary:** scenarios run on Linux containers; real constrained hardware, bootloaders, RTOS firmware, and physical interfaces introduce attack surfaces not covered here.

- **Controller trust:** official claims require controller-side verification, signed aggregate results, and auditable binding to the runner, model, budget, and suite manifest; worker-submitted logs alone are not a trust anchor.
- **Oracle boundaries:** matching contracts and bounded advisory categories still require expert review even though severity fallback has been removed.
- **Prompt and model scope:** results use a single prompt family; per-model prompt tuning may change absolute scores even if the architecture trend persists.
- **Scenario coverage:** the benchmark covers fixed topological patterns and vulnerabilities tractably injectable via Ansible; dynamic deployments, RF-layer attacks, and supply-chain weaknesses are out of scope.

9. Conclusion

Single-host benchmarks cannot capture the protocol roles and reachability dependencies that shape IoT/OT risk. This paper introduces IoTChainBench, with 23 public development scenarios and six sealed confirmatory profiles, and LANCE, a six-phase network-native harness for discovery, verification, intrusion, and reporting.

The methodological contribution is a fail-closed separation of detection, verified exploitation, severity-weighted coverage, negative controls, logical dependencies, and enforced network pivots. The final empirical claim will be drawn only from the pre-committed 54-run blind sealed matrix; public informed-versus-blind runs remain diagnostic. We release the public scenarios, playbooks, ground truth, matching contracts, and evaluation code to support reproducible research in this setting.

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